PAPER_724: UQFF Quantum Variable Documents: delta_sw, kappa, P_SCm, v_sw, omega_c

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Abstract

Five quantum variable documents define the superwave and calibration parameters: δ_{sw} (superwave amplitude perturbation), κ (UQFF calibration decay constant), P_{SCm} (SCm core pressure normalization), v_{sw} (superwave velocity), and ω_c (slow UQFF cycle frequency). These parameters govern the U_{g2} superwave correction and the temporal evolution of all UQFF reaction energies.

1. Variable Registry

Variable	Value	Physical Role
δ_{sw}	0.01	U_g2 superwave perturbation
κ	5×10^{-4} day^{-1}	E_react decay calibration
P_{SCm}	1.0	SCm pressure normalization for U_m
v_{sw}	5×10^{5} m/s	Superwave speed (aether/solar wind analog)
ω_c	1.585×10^{-8} rad/s	$\mu_j(t)$ modulation frequency

2. Superwave Correction in U_g2

$$U_{g2} = k_2 \cdot \frac{(\rho_{UA} + \rho_{SCm}) M_s}{r^2} \cdot S(r - R_b) \cdot \underbrace{(1 + \delta_{sw} \cdot v_{sw})}_{\text{superwave}} \cdot H_{SCm} \cdot E_{react}$$

3. UQFF Calibration

$$E_{react}(t) = 10^{46} \cdot e^{-\kappa t}, \quad \kappa = 5 \times 10^{-4}\,\text{day}^{-1}$$

4. Slow-Cycle Modulation

$$\mu_j(t) = \left(10^3 + 0.4 \sin(\omega_c t)\right) \cdot \mu_J^{base}$$

References

- UQFF KB grok_share_ba508f76c8e.txt entry #74
- Session 176, v5.33

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